# HDPCR™ SARS-CoV-2 Assay

Instructions for Use – Version 8, December 2020

IVD

COVID-19 Emergency Use Authorization Only

For *in vitro* diagnostic (IVD) Use | Rx Only

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# HDPCR™ SARS-CoV-2 Assay

*In vitro* diagnostic Real-Time PCR test for Coronavirus COVID-19

| For Use With: | | |
|---|---|---|
| **Sample Types** | **Extraction Platforms** | **PCR Platforms** |
| Nasopharyngeal swabs oropharyngeal swabs, anterior nasal swabs, mid-turbinate nasal swabs, nasal aspirate, nasal wash, and bronchoalveolar lavage (BAL) specimens | Roche MagNA Pure 24<br><br>Thermo Scientific™ KingFisher™ Flex | Applied Biosystems™ 7500 Fast<br><br>Applied Biosystems™ 7500 Fast Dx<br><br>Applied Biosystems™ QuantStudio™ 5 (96-Well, 0.2 mL Block and 384-Well Block)<br><br>Applied Biosystems™ QuantStudio™ 7 (96-Well Fast Block and 384-Well Block)<br><br>Applied Biosystems™ QuantStudio™ 12K Flex (96-Well Fast Block and 384-Well Block) |

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# HDPCR™ SARS-CoV-2 Assay

For COVID-19 Emergency Use Authorization Only

Instructions for Use

## *Intended Use*

The HDPCR™ SARS-CoV-2 Assay is a reverse transcription real-time polymerase chain reaction (qRT-PCR) test intended for the qualitative detection of nucleic acid from SARS-CoV-2 in human nasopharyngeal swabs, oropharyngeal swabs, anterior nasal swabs, mid-turbinate nasal swabs as well as nasal aspirate, nasal wash, and bronchoalveolar lavage (BAL) specimens from individuals who are suspected of COVID-19 by their healthcare provider. Testing is limited to laboratories certified under the Clinical Laboratory Improvement Amendments of 1988 (CLIA), 42 U.S.C. §263a, to perform high complexity tests.

Results are for the identification of SARS-CoV-2 RNA. The SARS-CoV-2 RNA is generally detectable in respiratory specimens during the acute phase of infection. Positive results are indicative of the presence of SARS-CoV-2 RNA; clinical correlation with patient history and other diagnostic information is necessary to determine patient infection status. Positive results do not rule out bacterial infection or co-infections with other respiratory viruses. The agent detected may not be the definite cause of disease. Laboratories within the United States and its territories are required to report all results to the appropriate public health authorities.

Negative results do not preclude SARS-CoV-2 infection and should not be used as the sole basis for patient management decisions. Negative results must be evaluated in combination with clinical observations, patient history, and epidemiological information.

The HDPCR SARS-CoV-2 Assay is intended for use by qualified clinical laboratory personnel specifically instructed and trained in the techniques of real-time PCR and *in vitro* diagnostic procedures. The HDPCR SARS-CoV-2 is only for use under the Food and Drug Administration's Emergency Use Authorization.

## *Principles of Procedure*

The HDPCR SARS-CoV-2 Assay uses TaqMan® probe chemistry and proprietary analysis to allow qRT-PCR multiplexing within a single-well. Viral nucleic acid is extracted from human nasopharyngeal swabs, oropharyngeal swabs, anterior nasal swabs, mid-turbinate nasal swabs, nasal aspirate, nasal wash, and bronchoalveolar lavage (BAL) specimens using either the Roche MagNA Pure 24 or the Thermo Scientific KingFisher Flex. The product includes the same N1 and N2 oligonucleotide primer and probe sequences for the detection of the SARS-CoV-2 viral RNA and the human RNase P gene used in the CDC 2019-Novel Coronavirus (2019-nCoV) Real-Time RT-PCR Diagnostic Panel for Emergency Use Only, effective 3/15/2020. Alternate reporter and quencher dyes are used to consolidate the reaction into a single well. Additional materials in the HDPCR SARS-CoV-2 Assay include enzyme and buffer mixes, extraction and assay run controls, and calibrators

to ensure accurate results. The N1 target is in the FAM channel, the N2 target is in the VIC channel, and the RNase P internal control (RNase P (IC)) is in the Cy5 channel.

## Assay Layout and Controls

The HDPCR primer and probe formulation contains TaqMan probes at a reaction limiting concentration which, in combination with ChromaCode Calibrators, allows for the use of end-point fluorescence detection of targets. Five calibrators are run per qRT-PCR plate, including two no template, and three triple-target replicate calibrators. The median calibrator endpoint from all three positive calibrator wells is taken as the target endpoint for each of the color channels and used to scale and compare sample data. This also allows for unification of expected values across instruments, just as a Ct does in traditional real-time PCR. Interpretation of these results is described below:

The ChromaCode Cloud Software independently assesses if the SARS-CoV-2 targets amplified in a sample and if the RNase P (IC) passed or failed. To be detected, a target amplification must reach a scaled endpoint fluorescence that is closer to the positive calibrator for each channel than the negative calibrator for that channel– FAM channel for N1 and VIC channel for N2. If this value is not reached, the target is not represented as detected and listed as “not detected" in the generated report. The purpose of the RNase P (IC) is to confirm a negative sample result. If the endpoint value of the RNase P (IC) amplification is closer to positive calibrator in the Cy5 channel, the internal control passes. If its endpoint is closer to the negative calibrator in the Cy5 channel, and neither N1 or N2 are detected the internal control fails and the sample well is invalid. If its endpoint is not closer to the positive calibrator and either N1, N2, or both are detected, the internal control is not assessed.

The ChromaCode Cloud Software assesses run success of the positive run control the negative run control, and the SARS-CoV-2 Assay Calibrators. The positive run control, COV_Pos, contains an RNA transcript of the Nucleocapsid gene and is in a matrix with human DNA. This control is added directly to the master mix on the PCR plate. One COV_Pos is run per qRT-PCR run to verify the master mix was appropriately made, by confirming reverse transcriptase activity and the PCR amplification of N1, N2 and RNase P (IC). All targets should amplify in the COV_Pos well. If this control fails, any or all of the three targets did not amplify, the plate is invalid. The negative run control (NTC) is added directly to the master mix on the PCR plate. One NTC is run per qRT-PCR run to confirm that there is no contamination in the master mix or plate set up. No targets should amplify in the NTC well. If this control fails due to aberrant amplification, the plate is invalid.

A plate is marked as invalid if the SARS-CoV-2 Assay calibrators fail the established QC criteria.

The negative extraction control, COV_Neg, is human DNA in a stabilizing matrix. This control is processed like a specimen, as it goes through extraction and qRT-PCR to monitor for cross contamination and the successful extraction of nucleic acid. One COV_Neg is run in every unique extraction process represented on a SARS-CoV-2 run. This control must be manually interpreted. A successful COV_Neg would have amplification of only the RNase P (IC) target and no others. If the COV_Neg fails, the samples processed in the same extraction run also should be manually interpreted as failed.

An interpretation guide to the SARS-CoV-2 Assay Controls is found in Table 1.

*Table 1 HDPCR SARS-CoV-2 Assay Controls*

| Control: | Controls for: | Control Requirement: |
|---|---|---|
| NTC (No Template Control) | Contamination in master mix/plate set up | One per qRT-PCR Plate; Control Passed |
| COV_Pos (Positive Run Control) | qRT-PCR Process Control | One per qRT-PCR Plate; Control Passed |
| COV_Neg (Negative Extraction Control) | Extraction Control, qRT-PCR Process for RNase P | One Per Extraction; Control Passed |
| RNase P (in HDPCR Mix) in Negative Sample | Confirms full process for negative samples | Built in for all qRT-PCR Wells; Pass When Sample Negative |

## *Materials Provided and Storage*

The product is available in a low volume test configuration (PN: 0683) or a High Throughput (HT) test configuration (PN: 0904 and PN: 0905). In order to run the HDPCR SARS-CoV-2 Assay in an HT configuration, both PNs 0904 and 0905 are required.
Control kits (PN: 0690) are available to be ordered, and are required for both configurations

*Table 2 HDPCR SARS-CoV-2 Assay, 480 Tests (PN: 0683)*

| Item | Part Number | QTY | Vol, µL | Shipping Condition | Storage Condition |
|---|---|---|---|---|---|
| HDPCR SARS-CoV-2 Subkit (480 Tests): | 0682 | 1 Kit | N/A | 2-8°C | -25 to -15°C |
| Enzyme Mix 03E | 0688 | 5 Tubes | 1050 | 2-8°C | -25 to -15°C |
| HDPCR SARS-CoV-2 Mix | 0674 | 5 Tubes | 450 | 2-8°C | -25 to -15°C |
| Reverse Transcriptase 01 | 0081 | 5 Tubes | 120 | 2-8°C | -25 to -15°C |
| COV_A | 0675 | 2 Tubes | 80 | 2-8°C | -25 to -15°C |
| COV_B | 0676 | 2 Tubes | 80 | 2-8°C | -25 to -15°C |
| COV_C | 0677 | 2 Tubes | 80 | 2-8°C | -25 to -15°C |
| COV_D | 0678 | 2 Tubes | 80 | 2-8°C | -25 to -15°C |
| COV_E | 0679 | 2 Tubes | 80 | 2-8°C | -25 to -15°C |

*Table 3: HDPCR SARS-CoV-2 Box 1, HT Assay, 2592 Rxn (PN: 0904)*

| Item | Part Number | QTY | Vol, µL | Shipping Condition | Storage Condition |
|---|---|---|---|---|---|
| Enzyme Mix 03E | 0688 | 27 Tubes | 1050 | 2-8°C | -25 to -15°C |
| HDPCR SARS-CoV-2 Mix | 0674 | 27 Tubes | 450 | 2-8°C | -25 to -15°C |
| Reverse Transcriptase 01 | 0081 | 27 Tubes | 120 | 2-8°C | -25 to -15°C |

*Table 4: HDPCR SARS-CoV-2 Box 2, HT Calibrators, 36 Runs (PN: 0905)*

| Item | Part Number | QTY | Vol, µL | Shipping Condition | Storage Condition |
|---|---|---|---|---|---|
| COV_A | 0675 | 9 Tubes | 80 | 2-8°C | -25 to -15°C |
| COV_B | 0676 | 9 Tubes | 80 | 2-8°C | -25 to -15°C |
| COV_C | 0677 | 9 Tubes | 80 | 2-8°C | -25 to -15°C |
| COV_D | 0678 | 9 Tubes | 80 | 2-8°C | -25 to -15 °C |
| COV_E | 0679 | 9 Tubes | 80 | 2-8°C | -25 to -15 °C |

*Table 5 HDPCR SARS-CoV-2 Controls, 20 Extraction Runs (PN: 0690)*

| Item | Part Number | QTY | Vol, µL | Shipping Condition | Storage Condition |
|---|---|---|---|---|---|
| COV_Neg | 0681 | 20 Tubes | 200 | Dry ice | -25 to -15°C |
| HDPCR SARS-CoV-2 Pos Ctrl/Diluent, 4ea: | 0696 | 1 Kit | N/A | Dry Ice | -25 to -15°C |
| COV_Pos | 0680 | 4 Tubes | 70 | Dry ice | -25 to -15°C |
| Diluent | 0695 | 4 Tubes | 200 | Dry ice | -25 to -15°C |

HDPCR SARS-CoV-2 Assay and Controls Kits can be stored between -25 and -15°C for up to 12 months from date of manufacture. Assay and Control Kits should not be used beyond expiration date listed on labels.

The HDPCR SARS-CoV-2 Mix, Enzyme Mix 03E and Reverse Transcriptase 01 may be thawed and used up to 4 times, storing at -25 to -15°C between uses.

The COV_A, COV_B, COV_C, COV_D, and COV_E calibrators may be thawed and used up to 4 times, storing at -25 to -15°C between uses.

The COV_Pos may be thawed and used up to 4 times, storing at -25 to -15°C between uses.

The Diluent may be thawed and used up to 4 times, storing at -25 to -15°C between uses.

The COV_Neg is single use only and may not be refrozen.

# *Materials Required but Not Provided*

Equipment:

- One Extraction System

| Instrument | Catalog No. |
|---|---|
| Roche MagNA Pure 24 Nucleic Acid Extraction System | 07290519001 |
| Thermo Scientific™ KingFisher™ Flex Extraction System | 5400630 |

- One qRT-PCR Instrument

| Instrument | Software | Catalog No. |
|---|---|---|
| Applied Biosystems™ 7500 Fast | 2.3 | 4351106 |
| Applied Biosystems™ 7500 Fast Dx* | 1.4.1 | 4406984 |
| Applied Biosystems™ QuantStudio™ 5, 96-Well, 0.2 mL | 1.4.3 | A28569 |
| Applied Biosystems™ QuantStudio™ 5 Flex, 384-Well | 1.5.1 | A28140 |
| Applied Biosystems™ QuantStudio™ 7 Flex, 96-Well | 1.3 | 4485698 |
| Applied Biosystems™ QuantStudio™ 7 Flex, 384-Well | 1.3 | 4485695 |
| Applied Biosystems™ QuantStudio™ 12K Flex, 96-Well | 1.3 | 4471088 |
| Applied Biosystems™ QuantStudio™ 12K Flex, 384-Well | 1.2.2 | 4471134 |

*Note: Instruments must be in current calibration per manufacturer specifications. qRT-PCR instrumentation must be calibrated for the following dyes: FAM, VIC, and Cy5.*

**If using the Applied Biosystems™ 7500 Fast Dx, the user must also download 7500 Software v2.3 for run file conversion*

- Vortex mixer
- Mini centrifuge
- PCR plate centrifuge
- Pipettes for volumes 5 to 1000 µL

Consumables:

- Molecular Grade RNase/DNase Free Water (for No Template Control)
- DNase/RNase free, sterile, filter tips for volumes 5 to 1000 µL
- DNase/RNase free, sterile tubes
  - 2 mL tubes
- Disposable gloves

Specific Consumables:

| Consumable | Catalog No. |
|---|---|
| Applied Biosystems™ MicroAmp™ EnduraPlate™ Optical 96-Well Fast Clear Reaction Plate with Barcode | 4483485 |
| Applied Biosystems MicroAmp Optical 96-Well Reaction Plate (for Applied Biosystems™ QuantStudio™ 5 Flex Software Version 1.4.3; 96-Well, 0.2mL: Catalog Number A28569) | N8010560 |
| Applied Biosystems MicroAmp Optical 384-Well Reaction Plate with Barcode | 4326270 |
| Bio-Rad Hard-Shell® 384-Well PCR Plates (for Applied Biosystems™ QuantStudio™ 5 Flex Software Version 1.5.1; 384-Well: Catalog Number A28140) | HSP3805 |
| Applied Biosystems™ MicroAmp™ Optical Adhesive Film PCR/Real-time PCR Compatible | 4311971 |
| Roche MagNA Pure 24 Extraction System: | |
| Roche MagNA Pure 24 Total Nucleic Acid Isolation Kit | 07658036001 |
| Roche MagNA Pure 24 Tip Park/Piercing Tool | 07345585001 |
| Roche MagNA Pure 96 Sealing Foil | 06241638001 |
| Roche MagNA Pure Filter Tips 1000 µL | 06241620001 |
| Roche MagNA Pure 24 Processing Cartridge | 07345577001 |
| Thermo Scientific™ KingFisher™ Flex Extraction System | |
| Applied Biosystems™ MagMAX Viral/Pathogen Nucleic Acid Isolation Kit | A42352 |
| KingFisher 96 well accessory kit from Macherey Nagel | 744951 |

# *Warnings and Precautions*

There are no known hazardous substances included in the manufacture of the HDPCR SARS-CoV-2 Assay. Safety Data Sheets are available online at https://chromacodecloud.com/downloads or through ChromaCode Customer Support at customer.support@chromacode.com

Additional material or chemicals required for the use of the HDPCR SARS-CoV-2 Assay should be closely examined by the user. The user should carefully read all warnings, instructions or Safety Data Sheets provided by the supplier and follow the general safety precautions when handling biohazards, chemicals and other materials.

## General Precautions

- The HDPCR SARS-CoV-2 Assay is for *in vitro* diagnostic use (IVD) only. Rx Only.
- For use under COVID-19 Emergency Use Authorization Only.
- Standard precautions and procedures should be taken:
  - when handling and extracting human samples;
  - when using extraction instruments;
  - disposing of samples, extracted material and waste.

- This test has not been FDA cleared or approved; the test has been authorized by FDA under an Emergency Use Authorization (EUA) for use by laboratories certified under the Clinical Laboratory Improvement Amendments (CLIA) of 1988, 42 U.S.C. §263a, to perform high complexity tests.
- This test has been authorized only for the detection of nucleic acid from SARS-CoV-2, not for any other viruses or pathogens.
- This test is only authorized for the duration of the declaration that circumstances exist justifying the authorization of emergency use of in vitro diagnostic tests for detection and/or diagnosis of COVID-19 under Section 564(b)(1) of the Act, 21 U.S.C. § 360bbb-3(b)(1), unless the authorization is terminated or revoked sooner.
- Standard precautions and procedures should be taken when handling and extracting human samples.
- Standard precautions and procedures should be taken when using extraction instruments.
- Standard precautions and procedures should be taken when disposing of samples, extracted material and waste.
- Dispose of reagents according to local regulations.
- Do not use reagents after their recommended stability time frame.
- Do not mix reagent lots from different HDPCR SARS-CoV-2 Assay kits.
- Avoid contamination by following good laboratory practices, wearing proper personal protective equipment, segregating workflow, and decontaminating workspace appropriately.
- Ensure all consumables are DNase and RNase free.

## *Specimen Collection and Storage*

Upper respiratory and BAL specimens should be collected using standard procedures and recommendations from the collection device manufacturer. Swab specimens can be collected in UTM/VTM/Liquid Amies or equivalent. Specimens should not be collected in saline.

Please refer to the Interim Guidelines for Collecting, Handling, and Testing Clinical Specimens from Patients Under Investigation for the 2019 Novel Coronavirus (2019-nCoV) provided by the CDC, www.cdc.gov.

- Samples can be stored at 2-8°C for 72 hours after collection prior to extraction. If samples need be transported, maintain 2-8°C on ice packs for overnight shipment.
- For longer term storage, unextracted samples can be stored at ≤-70°C. If samples need be transported, maintain ≤-70°C on dry ice for overnight shipment.
- Extracted nucleic acids can be stored at ≤-70°C. If samples need be transported, maintain ≤-70°C for overnight shipment.

*Note: Specimens must be packaged, shipped, and transported according to the current edition of the International Air Transport Association Dangerous Goods Regulation. Follow shipping regulations for UN 3373 Biological Substance, Category B when sending potential 2019-nCoV specimens.*

# *Running a New Test*

## Preparing to Run Assay on Instrument for the First Time

Prior to starting runs on any new instrument, the instrument must be registered on ChromaCode Cloud using the instrument's serial number and the appropriate run template file for that instrument must be downloaded.

*Note: Any instrument running the HDPCR SARS-CoV-2 Assay must be calibrated for the following dyes: FAM, VIC, and Cy5*

### Registering an Instrument on ChromaCode Cloud

Once your institution has been given access to the ChromaCode Cloud, the administrator will be able to upload lab instrument(s) to analyze HDPCR data.

1. Open a new window in Google Chrome™, or other ChromaCode Cloud compatible browser, and navigate to https://chromacodecloud.com
2. Click the **Admin** link at the top of the page.
3. Select **Add Instrument** at the top right of the page.
   *Note: The user will need to have been designated as the administrator to be able to access this functionality.*
4. Select the instrument model type in the drop-down menu labeled **Model**.
5. Enter a desired lab nickname for the instrument.
6. Enter the instrument's serial number; the number can be found on the side of the instrument.
7. Select **Save Instrument**.
8. Your instrument should now be listed on the page with a green checkmark indicating *equalization not required*.

### Download the Template Run File

The Template File contains all the parameters preconfigured to run the HDPCR SARS-CoV-2 Assay, including the run parameters:

*Table 6 Thermal Cycling Conditions for HDPCR SARS-CoV-2 Assay*

| Stage | Temperature(°C): | Time: | Reps: |
|---|---|---|---|
| 1 | 50.0 | 15:00 | 1 |
| 2 | 95.0 | 2:00 | 1 |
| 3 | 95.0 | 0:03 | 55 |
| | 58.0* | 1:00 | |

*This step should be the optical read step

To download the Template Run File:

1. On ChromaCode Cloud, navigate to **Downloads**.

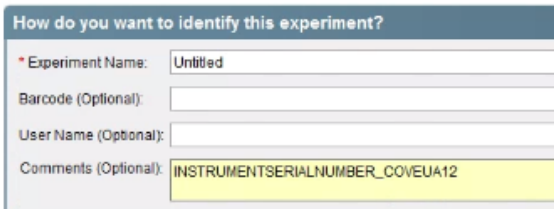
2. Download the appropriate .edt or .sdt template run file for your instrument type and save the template file on your instrument. *Note: Users need only download the template file and save to their instrument upon first use.*

*Table 7 Template Files for HDPCR SARS-CoV-2 Assay*

| Instrument | .edt/.sdt filename |
|---|---|
| Applied Biosystems™ 7500 Fast | COVEUA12_7500Fast_template |
| Applied Biosystems™ 7500 Fast Dx | COVEUA12_7500FastDx_template |
| Applied Biosystems™ QuantStudio™ 5 96-Well, 0.2mL | COVEUA12_QS5_96well_template |
| Applied Biosystems™ QuantStudio™ 5 384-Well | COVEUA12_QS5_384well_template |
| Applied Biosystems™ QuantStudio™ 7 96-Well | COVEUA12_QS7_96well_template |
| Applied Biosystems™ QuantStudio™ 7 384-Well | COVEUA12_QS7_384well_template |
| Applied Biosystems™ QuantStudio™ 12K Flex 96-Well | COVEUA12_QS12K_96well_template |
| Applied Biosystems™ QuantStudio™ 12K Flex 384-Well | COVEUA12_QS12K_384well_template |

### *Applied Biosystems™ 7500 Fast Template File*

1. Open the appropriate template, **COVEUA12_7500Fast_template** on your instrument.
2. Under the **Setup** tab on the left side, select **Experiment Properties**.
3. Add your instrument's serial number to the comments field where labeled
   a. Replace **InstrumentSerialNumber** on a 7500 Fast Instrument
   EX: 1234567890_COVEUA12




4. Select **File** on the Navigation tab and select **Save as Template**.
5. Name the file **COVEUA12_7500FAST_template.edt** and save in a preferred location.

### *Applied Biosystems™ 7500 Fast Dx, QuantStudio™ 12K Flex, QuantStudio™ 5 96-Well 0.2mL and 384-Well, and QuantStudio™ 7 96- and 384-Well Template Files*

1. Open the COVEUA12 template file on your instrument and proceed to running HDPCR SARS-CoV-2 Assay.

## Nucleic Acid Extraction

Refer to Roche MagNA Pure 24 or Thermo Scientific™ KingFisher™ Flex User Manual for full system usage and maintenance details. Use the following protocol and isolation kit depending on PCR instrument selected:

- Pathogen 200 2.0 Protocol and the MagNA Pure 24 Total Nucleic Acid Isolation Kit (Product Number 07658036001) on the Roche MagNA Pure 24 System
- MVP_Flex Protocol with the Applied Biosystems MagMAX Viral/Pathogen Nucleic Acid Isolation Kit (Product Number A42352) on the Thermo Scientific KingFisher Flex System

Overview:

1. Vortex primary specimen container to homogenize.
2. Add 200 µL of the specimen to extraction cartridge.
3. Extract in accordance with standard procedure, eluting into 50 µL.
4. Include one COV_Neg control on every extraction run, treating the control the same way as a specimen.
   *Note: Store COV_Neg at 4ºC once thawed and use within 1 day of thawing. This control is single use only and may not be refrozen.*
5. Store the extracted samples on cold blocks or ice if will be used immediately, otherwise freeze at ≤-70ºC.

## Create the Plate Layout Map

1. Open provided template file on thermal cycling instrument.
2. Assign a well for each of the samples.
3. Assign a well for each of the five HDPCR SARS-CoV-2 calibrators.

| | |
|---|---|
| COV_A | COV_D |
| COV_B | COV_E |
| COV_C | |

4. Assign a well for all controls: COV_Pos for the positive run control, COV_Neg for the negative extraction control(s), and NTC for the no template control.
   *Note: There must be one positive run control per qRT-PCR plate, one negative extraction control per extraction run, and one NTC per qRT-PCR plate.*
   *Note: If COV_Pos and NTC are not named exactly as such, the software will not recognize them as controls and will not interpret them accordingly.*
5. For the COV_Neg control, you may append the COV_Neg name to associate with a specific extraction run, as samples from more than one extraction can be run on the same plate.
6. Use your plate layout to load your samples, calibrators, and controls after preparing the amplification reaction mix.

## Prepare the Amplification Reaction Mix

*Note: Prepare the Amplification Reaction mix in a pre-PCR area.*

Thaw the following components (Table 8) at room temperature until no ice crystals remain:

*Table 8 Amplification Reaction Mix Components*

| Component |
|---|
| Enzyme Mix 03E |
| HDPCR SARS-CoV-2 Mix |
| Reverse Transcriptase 01 |

1. Vortex the HDPCR SARS-CoV-2 Mix for 5 seconds and spin to remove liquid from the cap.
2. Gently invert the Enzyme Mix 03E and Reverse Transcriptase 01 five (5) times and spin to remove liquid from the cap.
3. Prepare the Amplification Reaction Mix in a 2 mL tube according to the following table, where n = the number of reaction wells to be run.

   *Note: Remember to include all calibrators and controls in the calculation*

*Table 9 Amplification Reaction Mix Component Calculations*

| Component | Volume |
|---|---|
| Enzyme Mix 03E | (n+5) x 10 µL |
| HDPCR SARS-CoV-2 Mix | (n+5) x 4 µL |
| Reverse Transcriptase 01 | (n+5) x 1 µL |

4. Vortex Amplification Reaction Mix for 5 seconds and spin down to remove liquid from the cap.
5. Aliquot 15 µL of the amplification mix into each well that will be used for the run; use caution while loading to avoid introduction of bubbles into the well.

## Prepare COV_Pos Positive Control

*Note: Prepare COV_Pos Positive Control in a template positive area. The dilution of COV_Pos must be made fresh for each run of the HDPCR SARS-CoV-2 Assay and used within 1 hour of dilution. The dilution must be discarded after use.*

1. Thaw the COV_Pos control and the Diluent.
   *Note: Return the individual COV_Pos and Diluent tube to -20ºC if tubes will be used again. COV_Pos and Diluent may only be thawed and used up to 4 times.*
2. Vortex for 5 seconds and spin down to remove liquid from the cap.
3. Retrieve fresh, DNase/RNase free, sterile tube and add 45 µL of Diluent
4. Add 5 µL of COV_Pos to Diluent, creating a positive control of approximately a 100 copy/reaction.
5. Vortex for 5 seconds and spin down to remove liquid from the cap.

## Add Samples and Calibrators to Plate

*Note: Prepare and add samples and calibrators to plate in template positive area and keep samples on cold block or ice throughout plate set up*

1. Thaw the calibrators and the extracted samples (including extracted COV_Neg) if previously frozen.
2. Vortex for 5 seconds and spin down to remove liquid from the cap.
3. Add 5 µL of each calibrator to the well in accordance to the plate layout map.
   *Note: All 5 calibrators must be run on every plate*
4. Add 5 µL of the diluted COV_Pos and extracted COV_Neg controls to the wells in accordance to the plate layout map.
5. Add 5 µL of each sample to the wells in accordance to the plate layout map.
6. Add 5 µL of molecular grade water as the NTC in accordance to the plate layout map.
7. Place the film on top of the plate and use the squeegee to adhere the film, especially around the edges to avoid evaporation.
8. Spin the plate for 1 minute in a PCR plate spinner.

## Create a Run File and Start the Run

Refer to Instrument User Manuals for full system usage and maintenance details.

1. On instrument software, open the provided instrument specific template **SARS-CoV-2 template**.

*Note: At the start of each run, default comments on a 7500 Fast Dx Instrument must be replaced to have InstrumentSerialNumber_COVEUA12 (EX: 1234567890_COVEUA12)*

2. Ensure the Sample, Calibrator, and Control Names are correctly entered in software based on the Plate Layout Map.
3. Start the run in the software.

## ChromaCode Cloud Data Analysis

ChromaCode Cloud supports .eds and .xls file types from the supported instruments and software versions listed in the Materials Required section of this document. Customers must confirm as part of their laboratory assay validation process that the files generated from their instruments are compatible for upload to ChromaCode Cloud. Contact Technical Support with any questions on confirming ChromaCode Cloud file compatibility. The following provides specific instructions for saving and/or exporting data.

### ABI 7500 Fast, QuantStudio 5, QuantStudio 7 and QuantStudio 12k Flex

Ensure that .eds run files from instruments with a touch screen capability are saved directly from the computer attached to the instrument rather than the touchscreen monitor on the instrument.
The .eds file saved from the instrument computer can be uploaded for analysis.

Alternatively, an .xls file may be exported for analysis as follows:

1. Upon run completion, export the .xls run file by navigating to the **Export** tab, selecting .xls as **File Type**.
2. Select **Sample Setup, Raw Data**, **Amplification**, **Multicomponent**, and **Results.**
3. Select a **location** for the file to be saved and then select **Start Export.**
4. Upload the .xls file to ChromaCode Cloud for analysis.

### ABI 7500 Fast Dx

1. Once your run has completed on the 7500 Fast Dx instrument, save your .sds run file to a known location.
2. If necessary, transfer your .sds file to a computer that has the 7500 Software v2.3 installed.
3. Open the .sds file in the 7500 Software v2.3.
4. Confirm that the comments field contains the correct string (InstrumentSerialNumber_COVEUA12).
5. Navigate to **File, Save As**, and save the file with a .eds extension to a known location.

## Results Interpretation

SARS-CoV-2 N1, N2, and RNase P (IC) calls are determined by the ChromaCode Cloud Software.

### Upload the Run Data to ChromaCode Cloud

1. Open a new window in Google Chrome, or another ChromaCode Cloud compatible browser, and navigate to https://chromacodecloud.com
2. Log into ChromaCode Cloud.
3. Once on the home page, select **browse for instrument file to import**.

4. Browse to find the desired .eds or .xls run file to upload for analysis.
5. Select **Open** to begin analysis.

*Note: ChromaCode Cloud performs file integrity checks as part of the upload process. However, to reduce potential errors, upload file directly from instrument to ChromaCode Cloud. Do not open file on any additional software between completion of run and uploading to ChromaCode Cloud.*

## Plate Quality Control

*Note: The plate QC status and controls status must be assessed before sample result interpretation.*

The ChromaCode Cloud automatically assesses the plate QC status. This information is provided in a Plate Summary view and a Well Details page on the ChromaCode Cloud. Additionally, this information is consolidated in an exportable portable document format (pdf) report. The report can be downloaded from the Plate Summary page by clicking **view report** (seen in Figure 1).

*Figure 1 Plate Passing ChromaCode Cloud QC*

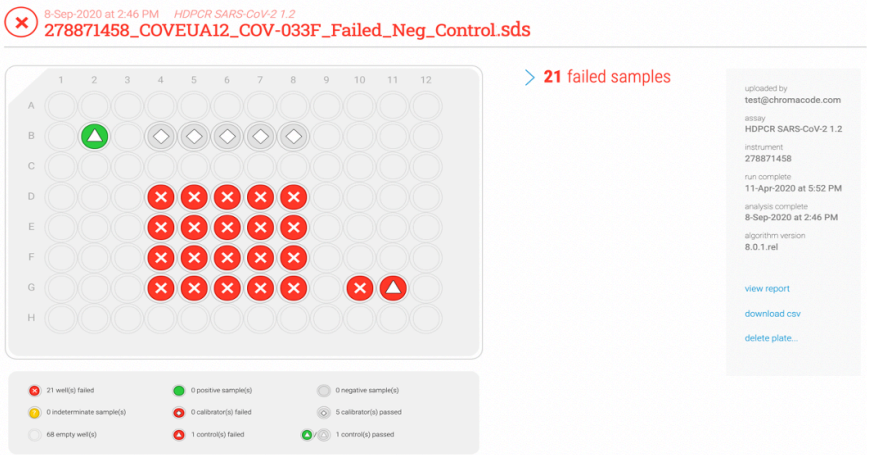



*Figure 2 Plate Summary Page with Failing Plate QC*

HDPCR SARS-CoV-2 calibrators must be run and pass quality control checks on every plate run. If these calibrators do not pass quality control checks, the plate run will be automatically flagged as failed on the ChromaCode Cloud by a red X next to the run name, which will be written in red text (an example of a failed plate in Figure 2). A plate that has passed plate QC will have a green check and the run name will be written in green text (Figure 1).

COV_Pos will be interpreted by the ChromaCode Cloud software, if named COV_Pos. If this control passes, it will be represented as a green well with a triangle shape. If this control fails, it will be represented as a red well with a triangle shape and the plate will fail.

The No Template Control (NTC) will be interpreted by the ChromaCode Cloud software if named NTC. If this control passes, it will be represented as a grey well with a triangle shape. If this control fails, it will be represented as a red well with a triangle shape and the plate will fail.

*Note: If the No Template Control (NTC) and Positive Run Control (COV_Pos) are named anything other than "NTC" and "COV_Pos", respectively, they will not be interpreted as part of the ChromaCode Cloud QC analysis.*

The COV_Neg control must be interpreted manually by the end user. To pass, this controls requires amplification of the RNase P (IC) and no amplification of the SARS-CoV-2 targets. The control passes if ChromaCode Cloud determines this well to be negative for the SARS-CoV-2 targets (viewed on the plate summary page or by generating a PDF results report).

A summary of the expected results for each control is summarized in the table below, along with action to take if a control does not have the expected results.

*Table 10 Plate QC Controls*

| Control | Passing Result | ChromaCode Cloud Text (Well Details Page) | Action if Fail |
|---|---|---|---|
| NTC<br>(No Template Control) | No Target Detection | control passed | Plate Invalid: repeat run with fresh NTC |
| COV_Pos<br>(Positive Run Control) | CoV-2 N1 detected<br>CoV-2 N2 detected<br>RNase P (IC) Passed | control passed | Plate Invalid: repeat run with fresh COV_Pos. |
| COV_Neg<br>(Negative Extraction Control) | CoV-2 N1 not detected<br>CoV-2 N2 not detected<br>RNase P (IC) Passed | no targets detected<br>Internal Control Passed | Sample Failure: for the samples run in the same extraction as the control. Repeat extraction for these samples and rerun. |

## Sample Results Interpretation

ChromaCode Cloud automatically assesses the amplification and detection of the N1 and N2 targets and the RNase P (IC) internal control status for each sample. This information is provided in a Plate Summary view (Figure 1) and a Well Details page. Additionally, this information is consolidated in an exportable report.

The end user must interpret the SARS-CoV-2 status of each sample based on N1, N2 and the RNase P (IC) status as presented in Table 11.

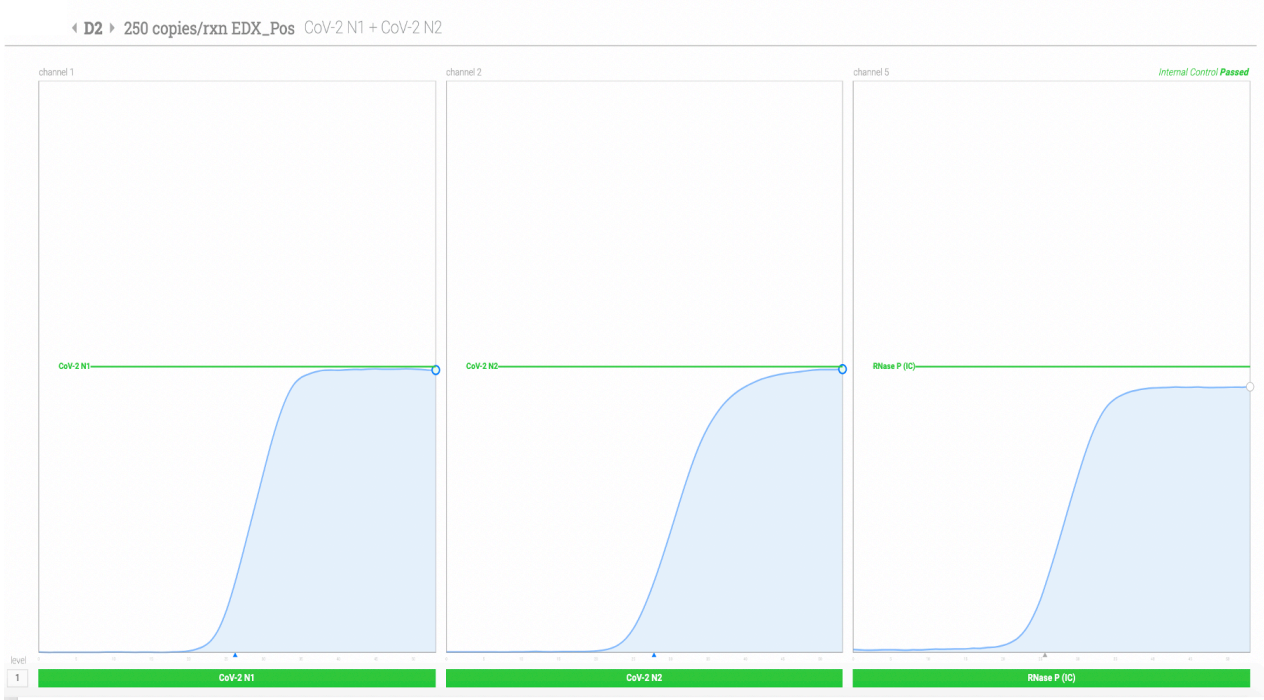



*Figure 3 Well Details Page Containing Targets*

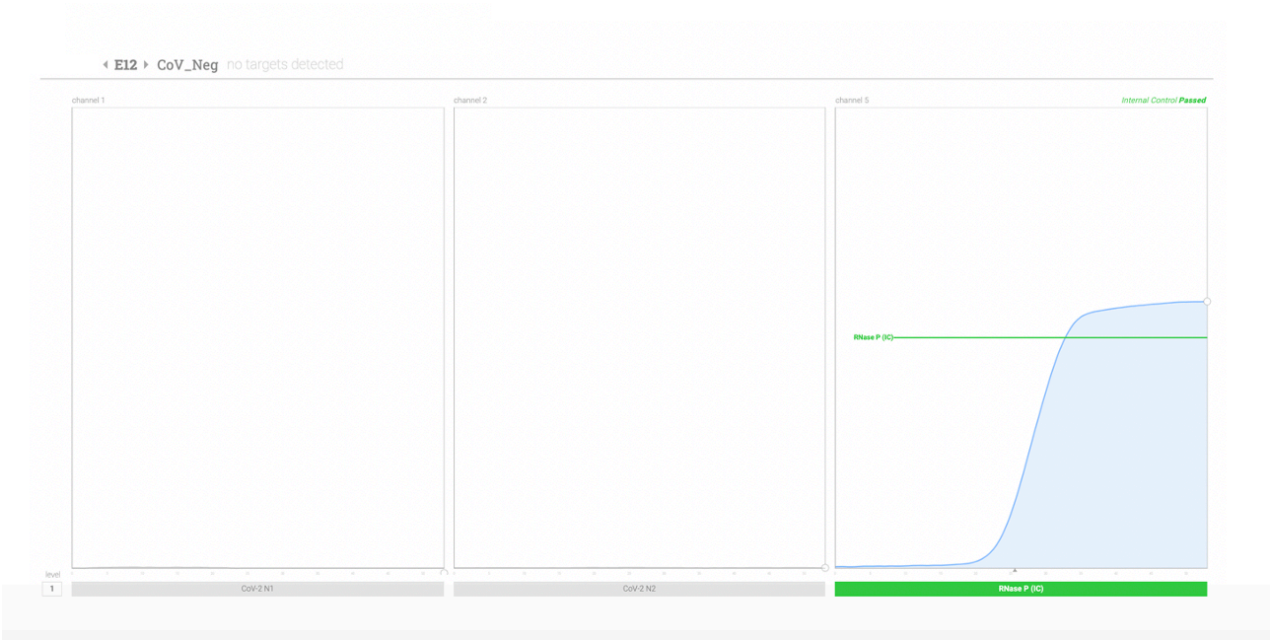



*Figure 4 Well Details Page Containing Negative Sample*

The Well Details page of any individual sample will highlight the target name listed under the channel in which that target is found in green if that target is detected. Additionally, detected targets (CoV-2 N1 and CoV-2 N2) are listed next to the sample name above the amplification curves (Figure 3). If there is no amplification, the target name listed under the channel will remain gray and there will be either only one target listed next to the sample name, or there will be "no targets detected" listed next to the sample name (Figure 4). There is text above the amplification curve of Channel 5 which indicates whether the Internal Control has succeeded by describing it as either "Internal Control Passed," "Internal Control Failed," or "Internal Control Not Assessed."

Reports can be generated through the ChromaCode Cloud.
The Run Summary report shows

- SARS-CoV-1 N1 or SARS-CoV-N2 when a target is detected
- NO TARGETS DETECTED when no targets are detected

The Sample Details Report is generated for every well and shows

- SARS-CoV-2 N1 as DETECTED or NOT DETECTED
- SARS-CoV-2 N2 as DETECTED or NOT DETECTED
- Human RNase P (IC) as PASSED, FAILED, or NOT ASSESSED

The following table (Table 11) outlines the expected results from the assay and potential recourse that must be taken.

*Table 11 HDPCR SARS-CoV-2 Results Interpretation*

| ChromaCode Analysis Output (Sample Detail Report) | | | Laboratory Interpretation & Actions | |
|---|---|---|---|---|
| SARS-CoV-2 N1 | SARS-CoV-2 N2 | Human RNase P (IC) | Report | Action |
| Detected | Detected | Passed or Not Assessed | SARS-CoV-2 Positive | Report result to appropriate health authorities. |
| Detected | Not Detected | Passed or Not Assessed | SARS-CoV-2 Presumptive Positive | Repeat testing of nucleic acid and/or re-extract and repeat HDPCR SARS-CoV-2. If the repeated result remains inconclusive, contact your State Public Health Laboratory or CDC for instructions for transfer of the specimen or further guidance. |
| Not Detected | Detected | Passed or Not Assessed | SARS-CoV-2 Presumptive Positive | Repeat testing of nucleic acid and/or re-extract and repeat HDPCR SARS-CoV-2. If the repeated result remains inconclusive, contact your State Public Health Laboratory or CDC for instructions for transfer of the specimen or further guidance. |
| Not Detected | Not Detected | Passed | SARS-CoV-2 Negative | Report result to appropriate health authorities. |
| Not Detected | Not Detected | Failed | Invalid Results | Repeat test, if second test result is invalid, report as invalid and recommend recollection if patient is still clinically indicated. |

## *Limitations*

The use of this assay as an *in vitro* diagnostic under the FDA COVID-19 Emergency Use Authorization (EUA) is limited to laboratories that are certified under the Clinical Laboratory Improvement Amendments of 1988 (CLIA), 42 U.S.C. § 263a, to perform high complexity test by Rx only.

Use of this assay is limited to personnel who are trained in the procedure. Failure to follow these instructions may lead to erroneous results.

The performance of *HDPCR SARS-CoV-2 Assay* was established using Nasopharyngeal Swab specimen type collected in UTM or VTM or Liquid Amies transport media. Nasal swabs, oropharyngeal swabs, mid-turbinate nasal swabs, nasal aspirate, nasal wash and BAL specimens are also considered acceptable specimen types for use with the HDPCR SARS-CoV-2 Assay but performance has not been established.

Samples must be collected according to manufacturer recommended protocols and transported and stored as described herein.

Samples should not be collected in saline.

The HDPCR SARS-CoV-2 Assay performance was established using the Roche MagNA Pure 24 System with the Roche MagNA Pure 24 Total Nucleic Acid Isolation Kit and the Pathogen 200 2.0 Protocol with the ABI QuantStudio 7 (96-Well and 384-Well), ABI 7500 Fast Dx, and ABI 7500 Fast.

Performance on the ABI QuantStudio 12K Flex (96-Well and 384-Well) and QuantStudio 5 (96-Well 0.2mL and 384-Well) were established using the Thermo Scientific KingFisher Flex and the Applied Biosystems MagMAX Viral/Pathogen Nucleic Acid Isolation Kit with the MVP_Flex protocol. Other extraction instrumentation and kits have not been tested with this assay.

All instrumentation used with the HDPCR SARS-CoV-2 Assay kits must be up to date on normal preventative maintenance and servicing/calibration schedules, including calibration with the FAM, VIC and Cy5 fluorescent dyes on the ABI 7500 Fast, ABI 7500 Fast Dx, QuantStudio 5, QuantStudio 7 Flex and QuantStudio 12K Flex real time PCR systems.

The effects of interfering substances have not been assessed with the HDPCR SARS-CoV-2 Assay.

The effect of vaccines, antiviral therapeutics, antibiotics, chemotherapeutic or immunosuppressant drugs have not been evaluated.

False-positive results may arise from various reasons, including, but not limited to the following:

- Contamination during specimen collection, handling, or preparation
- Contamination during assay preparation
- Incorrect sample labeling

False-negative results may arise from various reasons, including, but not limited to the following:

- Improper sample collection or storage
- Degradation of SARS-CoV-2 RNA
- Presence of inhibitory substances

- Use of extraction reagents or instrumentation not approved with this assay
- Incorrect sampling window
- Failure to follow instructions for use
- Mutations In SARS-CoV-2 target sequences

Nucleic acid may persist even after the virus is no longer viable.

Positive results must be reported to appropriate public health authorities, following state and national guidelines.

Negative test results do not exclude possibility of exposure to or infection with SARS-CoV-2 virus. Patient handling will be directed by healthcare professionals.

**Conditions of Authorization for the Laboratory**
The HDPCR SARS-CoV-2 Assay Letter of Authorization, along with the authorized Fact Sheet for Healthcare Providers, the authorized Fact Sheet for Patients, and authorized labeling are available on the FDA website: https://www.fda.gov/medical-devices/coronavirus-disease-2019-covid-19-emergency-use-authorizations-medical-devices/vitro-diagnostics-euas

However, to assist clinical laboratories running the HDPCR SARS-CoV-2 Assay, the relevant Conditions of Authorization are listed below:
A. Authorized laboratories[1] using the HDPCR SARS-CoV-2 Assay will include with test result reports, all authorized Fact Sheets. Under exigent circumstances, other appropriate methods for disseminating these Fact Sheets may be used, which may include mass media.

B. Authorized laboratories using the HDPCR SARS-CoV-2 Assay will use the HDPCR SARS-CoV-2 Assay as outlined in the HDPCR SARS-CoV-2 Assay Instructions for Use. Deviations from the authorized procedures, including the authorized instruments, authorized extraction methods, authorized clinical specimen types, authorized control materials, authorized other ancillary reagents and authorized materials required to perform the HDPCR SARS-CoV-2 Assay are not permitted.

C. Authorized laboratories that receive the HDPCR SARS-CoV-2 Assay must notify the relevant public health authorities of their intent to run the test prior to initiating testing.

D. Authorized laboratories using the HDPCR SARS-CoV-2 Assay will have a process in place for reporting test results to healthcare providers and relevant public health authorities, as appropriate.

E. Authorized laboratories will collect information on the performance of the test and report to DMD/OHT7-OIR/OPEQ/CDRH (via email: CDRH-EUA-Reporting@fda.hhs.gov) and ChromaCode Inc. local technical support center (via email: technical.support@chromacode.com) any suspected occurrence of false positive or false negative results and significant deviations from the established performance characteristics of the test of which they become aware.

F. All laboratory personnel using the test must be appropriately trained in PCR techniques and use appropriate laboratory and personal protective equipment when handling this kit, and use the test in accordance with the authorized labeling.

G. ChromaCode, Inc, authorized distributors, and authorized laboratories using the HDPCR SARS-CoV-2 Assay will ensure that any records associated with this EUA are maintained until otherwise notified by FDA. Such records will be made available to FDA for inspection upon request.

[1] The letter of authorization refers to, "Laboratories certified under the Clinical Laboratory Improvement Amendments of 1988 (CLIA), 42 U.S.C. §263a, to perform high complexity tests" as "authorized laboratories."

# *Performance Evaluation*

The following data demonstrate the performance of the HDPCR SARS-CoV-2 Assay. All sample extractions for samples used 200 µL of specimen input eluting into 50 µL. The Roche MagNA Pure 24 Pathogen 200 2.0 Protocol was used with the Total Nucleic Acid Isolation Kit with qRT-PCR on the Applied Biosystems QuantStudio 7 (96-Well and 384-Well), 7500 Fast Dx, and 7500 Fast instruments. The KingFisher MVP_Flex Protocol was used with the Fisher MagMAX Viral/Pathogen Nucleic Acid Isolation Kit with qRT-PCR on the Applied Biosystems QuantStudio 12K Flex (96-Well and 384-Well) Instrument and QuantStudio 5 (96-Well 0.2mL and 384-Well) Instruments.

## Analytical Sensitivity: Limit of Detection (LoD)

The LoD of the HDPCR SARS-CoV-2 Assay was determined using a two-stage approach. In the first stage, a preliminary LoD was established by testing either 3 or 5 replicates at each selected serial dilution. The lowest concentrations to detect either 3/3 or 5/5 replicates were moved to further evaluation in Stage 2, where the LoD was confirmed on each instrument by testing 20 replicates. Both Stage 1 and Stage 2 LoD testing used Armored RNA Quant® SARS-CoV-2 control obtained from Asuragen (Catalog Number 52030) spiked into negative nasopharyngeal swab specimens at designated concentrations. The results of this Stage 2 confirmation are seen in Table 12, indicating the established LoD for each qPCR instrument tested.

Results of LoD confirmatory testing, shown in Table 12, demonstrated an assay LoD (when both N1 and N2 targets are detected for at least 95% of sample replicates) of 1000 copies/mL for the Applied Biosystems 7500 Fast, Applied Biosystems QuantStudio 5 (96-Well 0.2mL), Applied Biosystems QuantStudio 5 (384-Well), Applied Biosystems QuantStudio 12K (384-Well), and the Applied Biosystems QuantStudio 7 (96-Well) systems, 250 copies/mL for the Applied Biosystems QuantStudio 12K (96-Well) system, and 2,000 copies/mL for the Applied Biosystems 7500 Fast Dx and the Applied Biosystems QuantStudio 7 (384-Well) systems.

*Table 12: Limit of Detection Confirmation Testing Summary*

| Instrument | Concentration (copies per mL) in extraction | *Concentration (copies per Reaction) | Detection Rate: CoV-2 N1 | Detection Rate: CoV-2 N2 |
|---|---|---|---|---|
| ABI 7500 Fast | **1000** | **20** | **20/20** | **20/20** |
| | 500** | 10 | 18/20 | 20/20 |
| ABI 7500 Fast Dx | **2000** | **40** | **20/20** | **20/20** |
| ABI QuantStudio 5 (96-Well 0.2 mL Block) | **1000** | **20** | **20/20** | **20/20** |
| ABI QuantStudio 5 (384-Well Block) | 2000 | 40 | 20/20 | 20/20 |
| | **1000** | **20** | **19/20*** ** | **19/20*** ** |
| ABI QuantStudio 7 (96-Well Block) | **1000** | **20** | **19/20** | **20/20** |
| | 500** | 10 | 18/20 | 20/20 |
| ABI QuantStudio 7 (384-Well Block) | **2000** | **40** | **20/20** | **20/20** |
| ABI QuantStudio 12K (96-Well Block) | 500 | 10 | 20/20 | 20/20 |
| | **250** | **5** | **20/20** | **19/20** |
| ABI QuantStudio 12K (384-Well Block) | **1000** | **20** | **20/20** | **20/20** |
| | 500 | 10 | 17/20 | 19/20 |

Bolded line represents final LoD for each instrument and block configuration
* Assumes 100% extraction efficiency
**Preliminary LoD Estimation Studies showed at 250 copies/mL 1/3 N1 and 3/3 N2 targets were detected. At 125 copies/mL 1/3 N1 and 2/3 N2 targets were detected. Confirmatory testing was not performed at these concentrations.
*** One well failed upon first test giving an invalid result, this sample was rerun as per the instructions for use and both N1 and N2 were detected in the repeated test

## Inclusivity: Analytical Sensitivity

The product includes the same N1 and N2 oligonucleotide primer and probe sequences for the detection of the SARS-CoV-2 viral RNA and the human RNase P gene used in the CDC 2019-Novel Coronavirus (2019-nCoV) Real-Time RT-PCR Diagnostic Panel for Emergency Use Only, effective 3/15/2020. The inclusivity analysis can be found in the referenced document found at the following URL:
https://www.fda.gov/media/134922/download

## Cross-Reactivity: Analytical Specificity

The product includes the same N1 and N2 oligonucleotide primer and probe sequences for the detection of the SARS-CoV-2 viral RNA and the human RNase P gene used in the CDC 2019-Novel Coronavirus (2019-nCoV) Real-Time RT-PCR Diagnostic Panel for Emergency Use Only, effective 3/15/2020. The exclusivity analysis can be found in the referenced document found at the following URL:
https://www.fda.gov/media/134922/download

# *Clinical Performance Evaluation*

The Clinical Evaluation of the HDPCR SARS-CoV-2 Assay was conducted using 30 negative nasopharyngeal swab specimens and 30 contrived positive samples. The contrived specimens were prepared by spiking negative nasopharyngeal specimens with varying concentrations of Armored RNA Quant® SARS-CoV-2 Control obtained from Asuragen (52030). Contrived specimens were extracted using the Roche MagNA Pure 24 Pathogen 200 2.0 Protocol with the Total Nucleic Acid Isolation Kit for qRT-PCR on the Applied Biosystems QuantStudio 7 (96-Well) and 7500 Fast instruments. The KingFisher MVP_Flex Protocol was used with the Fisher MagMAX Viral/Pathogen Nucleic Acid Isolation Kit for qRT-PCR on the Applied Biosystems QuantStudio 12K Flex (96-Well). For all extractions, 200 µL of contrived specimen input was eluted into 50 µL.

Twenty of the contrived reactive specimens were spiked with 1,000 copies/mL, and the remaining 10 were spiked in duplicate at 5,000, 25,000, 250,000, 2,500,000 and 25,000,000 copies/mL on the Applied Biosystems QuantStudio 7 (96-Well) and 7500 Fast instruments. Twenty of the contrived reactive specimens were spiked at 1000 copies/mL, and the remaining 10 were spiked in duplicate at 5,000, 10,000, 25,000, 50,000 and 500,000 copies/mL on the Applied Biosystems QuantStudio 12K Flex (96-Well) instrument. All instruments had 30 negative samples run in addition to contrived reactive specimens.

*Table 13 Clinical Evaluation Data Summary on Applied Biosystems 7500 Fast*

| Spiked RNA Concentration (copies/mL) | SARS-CoV-2 N1 | SARS-CoV-2 N2 | Presumptive Positive |
|---|---|---|---|
| Negative | 29/30 | 30/30 | 1/30* |
| 1,000 | 20/20 | 20/20 | 0/20 |
| 5,000 | 2/2 | 2/2 | 0/2 |
| 25,000 | 2/2 | 2/2 | 0/2 |
| 250,000 | 2/2 | 2/2 | 0/2 |
| 2,500,000 | 2/2 | 2/2 | 0/2 |
| 25,000,000 | 2/2 | 2/2 | 0/2 |

***Positive Percent Agreement: 30/30 = 100% (CI: 88.7-100%)***
***Negative Percent Agreement: 29/30 = 96.7% (CI: 83.3-99.4%)***
* A second extraction of this specimen was negative for both N1 and N2

*Table 14 Clinical Evaluation Data Summary on Applied Biosystems QuantStudio 7*

| SARS-CoV-2 Concentration (copies/mL) | SARS-CoV-2 N1 | SARS-CoV-2 N2 | Presumptive Positive |
|---|---|---|---|
| Negative | 29/30 | 30/30 | 1/30* |
| 1,000 | 20/20 | 20/20 | 0/20 |
| 5,000 | 2/2 | 2/2 | 0/2 |
| 25,000 | 2/2 | 2/2 | 0/2 |
| 250,000 | 2/2 | 2/2 | 0/2 |
| 2,500,000 | 2/2 | 2/2 | 0/2 |
| 25,000,000 | 2/2 | 2/2 | 0/2 |

***Positive Percent Agreement: 30/30 = 100% (CI: 88.7-100%)***
***Negative Percent Agreement: 29/30 = 96.7% (CI: 83.3-99.4%)***
* A second extraction of this specimen was negative for both N1 and N2

*Table 15 Clinical Evaluation Data Summary on Applied Biosystems QuantStudio 12K Flex*

| SARS-CoV-2 Concentration (copies/mL) | SARS-CoV-2 N1 | SARS-CoV-2 N2 | Presumptive Positive |
|---|---|---|---|
| Negative | 30/30 | 30/30 | 0/30 |
| 1,000 | 20/20 | 20/20 | 0/20 |
| 5,000 | 2/2 | 2/2 | 0/2 |
| 10,000 | 2/2 | 2/2 | 0/2 |
| 25,000 | 2/2 | 2/2 | 0/2 |
| 50,000 | 2/2 | 2/2 | 0/2 |
| 500,000 | 2/2 | 2/2 | 0/2 |

***Positive Percent Agreement: 30/30 = 100% (CI: 88.7-100%)***
***Negative Percent Agreement: 30/30 = 100% (CI: 88.7-100%)***

Clinical Evaluation of the HDPCR SARS-CoV-2 Assay was conducted on both the 384 and 96-Well format on the Applied Biosystems QuantStudio 7 instrument. The specimens were enrolled using the CDC 2019-Novel Coronavirus (2019-nCoV) Real-Time RT-PCR Diagnostic Panel with Promega Maxwell® extraction chemistry and run on the ABI QuantStudio Dx instrument. Specimens (NP swabs and combo NP/OP swabs collected in VTM) to be run on the HDPCR SARS-CoV-2 Assay were extracted using the Roche MagNA Pure 24 Pathogen 200 2.0 Protocol with the Total Nucleic Acid Isolation Kit. The results of these clinical specimens are seen in Table 16 and Table 18 for the 384-Well and 96-Well, respectively.

*Table 16 Clinical Evaluation QuantStudio 7, 384-Well Instrument (All enrolled samples)*

| Comparator Result | Total | Correct | Incorrect | Invalid | Presumptive Positive | Percent Agreement | 95% CI |
|---|---|---|---|---|---|---|---|
| Positive | 67 | 60 | 5 (FN) | 1 | 1 | 89.6% | 79.6-95.7% |
| Negative | 84 | 80 | 4 (FP) | 0 | 0 | 95.2% | 88.2-98.7% |

Based on the comparator method Ct values used for gathering the data in Table 16, 3 of the 5 false negatives and the presumptive positive are samples near or below the LoD for the comparator assay. Ct values yielded from the comparator method for these false negatives and the presumptive positive are seen below in Table 17.

*Table 17 Clinical Comparator Data (CDC Assay) for False Negative and Presumptive Positive Calls*

| Sample Identification | CDC Comparator N1 Ct Value | CDC Comparator N2 Ct Value |
|---|---|---|
| COV060 | 39 | 37 |
| COV064 | 35 | 36 |
| COV070* | 38 | 40 |
| COV107 | 36.7 | 39.6 |
| COV143 | 37.6 | 39.9 |
| COV150 | 34.8 | 37.2 |

*Presumptive positive with the HDPCR SARS-CoV-2 Assay

*Table 18 Clinical Evaluation QuantStudio 7, 96-Well Instrument*

| Comparator Result | Total | Correct | Incorrect | Invalid | Presumptive Positive | Percent Agreement | 95% CI |
|---|---|---|---|---|---|---|---|
| Positive | 39 | 37 | 2 (FN) | 0 | 0 | 94.9 | 82.7 - 99.4 |
| Negative | 41 | 39 | 2 (FP) | 0 | 0 | 95.1 | 83.4 – 99.4 |

## FDA SARS-CoV-2 Reference Panel Testing

The evaluation of sensitivity and MERS-CoV cross reactivity was performed using reference material (T1), blinded samples and a standard protocol provided by the FDA. The study included a range finding study and a confirmatory study for LoD. Blinded sample testing was used to establish specificity and to confirm the LoD. The extraction method and the instrument used were the Roche MagNA Pure 24 Pathogen 200 2.0 Protocol with the Total Nucleic Acid Isolation Kit with qRT-PCR on the Applied Biosystems 7500 Fast instrument. The results are summarized in Table 19.

*Table 19 Summary of LoD Confirmation Results using the FDA SARS-CoV-2 Reference Panel*

| Reference Materials Provided by the FDA | Specimen Type | HDPCR SARS-CoV-2 Assay LoD | Cross-Reactivity |
|---|---|---|---|
| SARS-CoV-2 | NPS in VTM | $5.4 \times 10^3$ NDU/mL | N/A |
| MERS-CoV | | N/A | ND |

NDU/mL: RNA NAAT detectable units/mL
N/A: Not applicable
ND: Not detected

# Document Revision History

Document ID 334738

| Revision | Effective Date | Description |
|---|---|---|
| 1 | 21 April 2020 | Original IFU pre-sub to FDA |
| 2 | 29 April 2020 | Updated to include the following instrumentation<br>• Applied BioSystems QuantStudio 12K qPCR instrument<br>• Thermo Fisher KingFisher extraction platform |
| 3 | 10 June 2020 | • Updated to include final Emergency Use Authorization |
| 4 | 11 September 2020 | • Inclusion of FDA SARS-CoV-2 Reference Panel Testing |
| 5 | 18 September 2020 | Expansion to include the following qPCR instruments:<br>• Applied BioSystems QuantStudio 12K Flex 384-Well<br>• Applied BioSystems QuantStudio 7 384-Well<br>• Applied BioSystems QuantStudio 5 384-Well<br>• Applied BioSystems QuantStudio 5 96-Well (0.2 mL)<br>• Applied BioSystems 7500 Fast Dx<br>Inclusion of Liquid Amies Transport Media |
| 6 | 09 November 2020 | • Applied BioSystems 7500 Fast Dx .sds file export workflow updated<br>• Language added regarding data file validation for ChromaCode Cloud |
| 7 | 08 December 2020 | • Inclusion of Clinical Evaluation on QuantStudio 7 96-Well instrument<br>• Inclusion of saline limitation |
| 8 | 15 December 2020 | Current Revision<br>• Inclusion of High Throughput (HT) Kit configuration option |

## Trademarks

HDPCR™ is a trademark of ChromaCode, Inc.
All other product names and trademarks are the property of their respective owners.

## Explanation of Symbols

The following symbols are present on the HDPCR™ SARS-CoV-2 Assay labels.

| Symbol | Definition |
|---|---|
| | Manufactured By |
| | Upper and Lower Storage Temperature Limitation |
| $R_x$ | By Prescription Only |
| LOT | Lot Number |
| QTY | Quantity |

| Symbol | Definition |
|---|---|
| REF | Product Catalog Number |
| | Use by Date |
| IVD | In vitro Diagnostic Medical Device |
| | Information |

## Manufacturing and Distribution Information

**Manufactured by**
ChromaCode, Inc.
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